

RSD of intermediate precision

$$RSD_{IP} = \frac{s_{IP}}{\bar{Z}} \times 100$$

RSD of reproducibility

$$RSD_R = \frac{s_R}{\bar{Z}} \times 100$$

For instance, the *RSD* of repeatability and reproducibility obtained from the LEAD dataset are:

$$RSD_r = 100 \times \frac{s_r}{\bar{Z}} = 100 \times \frac{0.0867}{2.047} = 4.23\%$$

$$RSD_R = 100 \times \frac{s_R}{\bar{Z}} = 100 \times \frac{0.1642}{2.047} = 8.02\%$$

All these notations are not equivalent but are considered convenient to characterize the precision of a method because they can be used to compare different methods, whatever the concentration range. Therefore, they are intensively used to establish regulatory validation criteria, as explained in Section 7.5.1. For example, an acceptance criterion for reproducibility is based on Horwitz's model, which was developed to decide whether an observed RSD_R value is acceptable or not.

It must be noted that computed *RSD* results from the ratio of two random variables, giving a new parameter with high uncertainty. In the case of method validation procedures, measurements are preferably done on special validation material, the assigned or reference value of which is known. It is then possible that the *RSD* denominator should be replaced by this value. The new ratios computed in this way are no more *RSD*, in the classical definition. They are sometimes considered as estimates of precision, regardless of the estimated trueness [11].

3.2.2.2 Variance of the Grand Mean

The method accuracy profile (MAP) is a method validation procedure presented in Chapter 5. All calculations are based on data collected under intermediate precision conditions and the former equations are adequate to compute the grand mean, standard deviations of repeatability, and intermediate precision. But it also requires an estimate of the variance of the grand mean. This additional parameter is given by Eq. (3.21). It depends on the variance ratio, noted A (see Eq. 3.24), which plays a fundamental role in the MAP procedure, as explained further.

Variance of the grand mean

$$s_{\bar{Z}}^2 = s_{IP}^2 \times \frac{1}{IJQ} \quad (3.21)$$

Standard deviation of the grand mean

$$s_{\bar{Z}} = s_{IP} \sqrt{\frac{1}{IJQ}} \quad (3.22)$$

Coefficient Q

$$Q = \frac{A + 1}{J \times A + 1} \quad (3.23)$$

Variance ratio

$$A = \frac{s_B^2}{s_r^2} \quad (3.24)$$

In Section 7.2, a procedure is also proposed to derive the MU from data collected under intermediate precision condition. This is recognized as a very promising solution to obtain MU.

3.3 Balanced and Unbalanced Experimental Design

According to the ANOVA vocabulary, the experimental design is said *to be balanced* when the number of replicates is the same at all levels of the factor (i.e. laboratories or series). For instance, the LEAD experimental design is balanced because all participants did the same number of replicates $J = 3$. The Figure 3.5 provides a convenient graphical representation of balanced and unbalanced experimental designs. The main rectangle represents the whole dataset, and small rectangles represent laboratories or measurement series. The height is proportional to the number of laboratories I and the width to the maximum number of replicates J . If the design is balanced, all gray rectangles are the same length. If the design is unbalanced, the length varies with the number of replicates n_i .

The statistical data processing described in Section 3.2 is only applicable to a balanced design. For unbalanced design, when all levels do not contain the same number of replicates, it is necessary to introduce n_i the number of replicates of laboratory/series i . The indexing of replicates is as follows: for the balanced design, it is $1 \leq j \leq J$, and for the unbalanced design, it is $1 \leq j \leq n_i$. Previous equations are consequently modified and become:

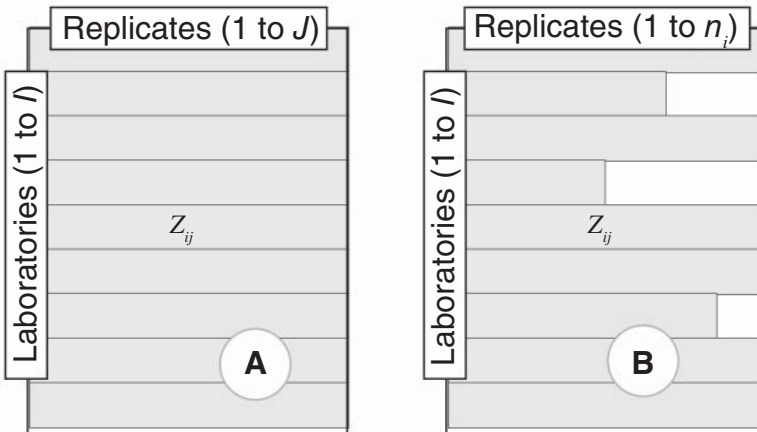


Figure 3.5 Graphical representation of the experimental design. (a) Balanced and (b) unbalanced.

Total number of measurements

$$N = \sum_i n_i \quad (3.25)$$

Corrected number of measurements

$$N' = N - \frac{\sum_i n_i^2}{N} \quad (3.26)$$

General ANOVA equation

$$\sum_i \sum_j (Z_{ij} - \bar{Z})^2 = \sum_i \sum_j (Z_{ij} - \bar{Z}_i)^2 + \sum_i n_i (\bar{Z}_i - \bar{Z})^2 \quad (3.27)$$

$$SS_t = SS_W + SS_B$$

Repeatability variance

$$s_r^2 = \frac{SS_W}{N - 1} \quad (3.28)$$

Between-series variance or interlaboratory variance

$$s_B^2 = \frac{(I - 1) \left(\frac{SS_B}{I - 1} - s_r^2 \right)}{N'} \quad (3.29)$$

These modifications are not of utmost importance, but interesting for software development. Obviously, when *RSD* are computed, they are modified in the same way as the standard deviation and grand mean are. Finally, the estimation of the precision parameters is more sophisticated than sometimes presented. Quite often, analysts claim they have obtained “a repeatability” by simply computing the average value $\bar{Z}$ and standard deviation s of 10 or 20 replicates performed on the same sample on the same day.

Although the measures are made under repeatability condition, this is a misunderstanding of the concept of repeatability as defined in the ISO 5725 standard. When replicates are simply performed on the same day, the total standard deviation may erroneously overestimate or underestimate the method’s precision. When the same experiment is repeated, it is frequent to obtain a different value. Therefore, it is compulsory to distribute replicates over several days under intermediate precision conditions, and extract the components of the total variance by applying the ANOVA algorithm to obtain a pertinent estimate of repeatability. Section 3.4 explains how to do this easily with classical worksheet software.

3.4 Software Implementation

3.4.1 ANOVA Classic Algorithm

Some care must be taken because most statistical software includes an ANOVA procedure. For example, the Excel “Analysis Toolpak” is an add-in containing the option “ANOVA: Single Factor.” This can be misleading because there are several ways to perform an ANOVA. In the Excel toolbox, it is assumed that the data is structured with a fixed effect factor.

Whereas for measurement method precision assessment, the model is assumed to be a *random effect factor*, as explained above. The algorithm for fixed effects is different and not adapted to the precision parameter estimation. Fixed effect ANOVA is devoted to making multiple comparisons of means as the factor values are controlled by the experimenter, such as different temperatures. While random factor ANOVA is used to extract the variance components of the dataset. The results obtained with the fixed effect model cannot be used to estimate the between-series or between-laboratory variance, and therefore the precision parameters. But it can be applied to verify if all laboratory means are statistically equal.

If there is a single mean that does not conform with the others, it is said to be an outlier, and the laboratory must be discarded. This is not the procedure implemented in the ISO 5725 standard, and a short presentation of outlier and straggler detection standardized methods is presented at the end of this chapter. Therefore, the Excel statistical add-in cannot be used for computing precision, and it is necessary to develop a specific tool to obtain useful results.

An example using the LEAD dataset is given in the Resource E worksheet. It is designed to manage 11 laboratories or series and 3 replicates/laboratories. It must be adapted to diverse situations where the numbers of participants (or series) and of replicates are larger. Some explanations may help in the implementation of this worksheet. As was the case for Resource A, the formulas are visible in column C and column B contains the expected result.

Raw data is input in area B4 : E14, one line par laboratory. The laboratory sum of squares is computed in column E with header “ SS_i .” The built-in statistical function DEVSQ allows one to directly obtain any sum of squared deviations. Fortunately, if cells are empty, they are considered absent and not put to 0. This means the same worksheet can be used when the number of series or replicates is smaller.

Due to the lack of space, corresponding formulas are not visible, but, for instance, in cell F4, this formula was input =DEVSQ (B4 : D4) , and the result was 0.003800. All SS_i are summed in cell B19 to give the SS_w . This is the direct application of the Eqs. (3.9) and (3.10) for the within-laboratories (or series) sum of squares. The total sum of squares SS_t is obtained in cell B20 by using the same function but applied to the whole data area =DEVSQ (B4 : D14) .

In cell B21, the calculation of SS_b by difference may pose a problem because the resulting value can be negative, which sometimes happens when all the means are almost equivalent. As explained, the ISO 5725 standard recommends forcing the value of s_L^2 (or s_b^2) to 0, otherwise, an error occurs when calculating the square root (a negative value having no real square root). The decision rule is simple:

$$\text{If } s_L^2 < 0 \rightarrow s_L^2 = 0$$

It is coded in two steps: first, compute s_L^2 in cell B21 by applying formula (3.13); then check if this value is strictly smaller than 0 with the Excel formula =IF (B23<0 ; 0 ; B23) . As explained later in this chapter, specialized statistical software offers another algorithm called restricted maximum likelihood (REML), which avoids this problem. As this worksheet is devoted to work on a balanced design, it is necessary to verify if this condition is satisfied.

The Excel built-in function COUNT, which returns the number of not empty cells, is used to make the control. The total number of non-null data is computed in cell B17 with =COUNT (B4 : D14) . The number of laboratories or series is reported in cell B16 with =COUNT (B4 : B14) and the number of replicates in cell F4 with =COUNT (B4 : E4) . The experimental design must be balanced, and the verification is done in cell B18 with the formula =IF (F4*B16<>B17; "Error"; F4) . This is a rather simplistic method of checking, as it may fail if the data layout is not correctly set up, but it is just to show what needs to be done to secure the template.

Resource E Precision parameters for a balanced design (Excel).

	A	B	C	D	E	F
1	Resource E: Precision parameters for a balanced design					
2	Lead mg/kg in mussel tissue					
3	Laboratories	Replicate. 1	Replicate. 2	Replicate. 3	ni	SSi
4	Lab 01	2.08	2.00	2.01	3	0.00380
5	Lab 02	2.00	1.93	1.89	3	0.00620
6	Lab 03	2.10	2.44	1.96	3	0.12187
7	Lab 04	2.45	2.34	2.49	3	0.01207
8	Lab 05	1.95	1.89	1.93	3	0.00187
9	Lab 06	1.85	1.91	1.89	3	0.00187
10	Lab 07	2.01	2.00	2.06	3	0.00207
11	Lab 08	2.00	2.09	1.98	3	0.00687
12	Lab 09	2.11	2.03	2.14	3	0.00647
13	Lab 10	2.02	1.98	1.97	3	0.00140
14	Lab 11	2.02	2.00	2.04	3	0.00080
15	Main parameters					
16	Number of laboratories (I)	11	=COUNT(B4:B14)			
17	Number of measurements (IJ)	33	=COUNT(B4:D14)			
18	Number of replicates (J)	3	=IF(E4*B16<>B17;"Error";E4)			
19	SSW	0.16527	=SUM(F4:F14)			
20	SSt	0.82385	=DEVSQ(B4:D14)			
21	SSB	0.65859	=B20-B19			
22	Repeatability variance (s²r)	0.00751	=B19/(B17-B16)			
23	Temporary between lab variance	0.01945	=((B21/(B16-1))-B22)/B18			
24	Between laboratory variance (s²L)	0.01945	=IF(B23<0;0;B23)			
25	Reproducibility variance (s²R)	0.02696	=B22+B24			
26	Precision					
27	Grand mean	2.047	=AVERAGE(B4:D14)			
28	Repeatability std. dev.	0.0867	=SQRT(B22)			
29	Between laboratory std. dev.	0.1395	=SQRT(B24)			
30	Reproducibility std. dev.	0.1642	=SQRT(B25)			
31	Limit of repeatability	0.2453	=2.83*B28			
32	Limit of reproducibility	0.4647	=2.83*B30			
33	Relative Std. Dev. Reproducibility (RSDR)	8.02%	=B30/B27			
34	Grand mean variance (s²Z)					
35	Variance ratio A	2.59	=B24/B22			
36	Coefficient Q	0.409	=(B35+1)/(B18*B35+1)			
37	Grand mean variance (s²Z)	0.0073	=B25*SQRT(1/(B17*B36))			
38	Grand mean std. dev. (sZ)	0.0856	=SQRT(B37)			

Finally, it is advisable to protect this worksheet against any unintentional changes. It should be remembered that the number of significant figures in a cell depends on its display format and not on the rounding of the data. If data rounding is necessary, use the ROUND function but always at the end of the computation. The case of an unbalanced dataset is addressed in the Section 3.4.2, and a modified